

workers' computer screens and into electronic mail; and the use of active or magic badges that can keep track of an employee's movements and locations. Increasingly, computers are being used to set tasks and performances for all levels of workers. Employers may view employees on closed-circuit TV; tap their phones, e-mail and network communications; and rummage through their computer files with or without employees' knowledge or consent—24 hours a day.

Electronic identification and monitoring are important tools within the management of animal husbandry systems. Ruminant activity is an indicator of animals' health status and production. Remote monitoring of animal behaviour in the environment can assist in managing both the animal and its environmental impact. GPS collars that record animal locations with high temporal frequency allow researchers to monitor both animal behaviour and interactions with the environment.

Tracking migrations is an important tool to better understand and protect species. Scientists today still attach tags, such as metal bands, to track animal movements. Metal bands require re-capture of animals for the scientists to gather data; data is thus limited to the animal's release and destination points.

Recent technologies have solved this problem. Electronic tags give scientists a complete, accurate picture of migration patterns. Some electronic tags give off repeating signals that are picked up by radio devices or satellites, while other electronic tags could include archival tags (or data loggers). Using RFID technology or satellites, scientists can track the location and movement of tagged animals without recapturing them. These electronic tags can provide a great deal of data.

Telemetry, in general, involves the use of a transmitter that is attached to an animal and sends out a signal in the form of radio waves, just as a



Electronic monitoring in sports

radio station does. Scientists may place the transmitter around an animal's ankle, neck, wing, carapace or dorsal fin. Alternatively, they may surgically implant it as internal radio transmitters, which have the advantage of remaining intact and functioning longer than traditional attachments, with protection from environmental variables and wear.

Geolocators or geologgers technology utilises a light sensor that tracks light-level data at regular intervals in order to determine a location based on the length of the day and the time of solar noon. Receivers can be placed in Earth-orbiting satellites, and networks (or groups) of satellites used to track animals. Each satellite in a network picks up electronic signals from a transmitter on an animal. Together, signals from all satellites determine the precise location of the animal. Satellites also track the animal's path as it moves. Satellite transmitters fitted to animals can also provide information about their physiological characteristics (temperature) and habitat use. Satellite tracking is useful especially because scientists do not have to follow the animal or recover its tag to get data on its whereabouts.

Satellite networks track the migration and territorial movements of caribou, sea turtles, whales, great white sharks, seals, elephants,

bald eagles, ospreys and vultures. Additionally, pop-up satellite archival tags are used on marine mammals and various species of fish. Thanks to these systems, conservationists can find the key sites for migratory species.

Uses of electronic monitoring

In sports. Present electronic monitoring systems use new and improved apparatus and method to control a television system, which includes a replay recorder, to monitor events of interest during sports contests. When the indicator detects an event of interest during the contest being monitored, such as an intrusion or the presence of an object along a line or boundary, a signal is sent to the video replay recorder (also known as an instant replay system) to continue recording for a predetermined interval and then cease.

In sports contests, several types of indicators assist officials in monitoring events of interest, particularly those along lines or boundaries—such as determining whether a tennis ball lands in or out of play during a tennis match. One type of indicator is electronic-laser indicator. Other types of line indicators include those utilising pressure-sensitive tapes as the line or boundary and those based on the detection of magnetic particles in the ball.